REal-Time Fractional Tracking (R-TFT): Recursive Resonance Fields in Patterned Media

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Abstract

This paper applies the Real-Time Fractional Tracking (R-TFT) framework to analyze patterned media such as metamaterials, photonic lattices, and structured surfaces. Unlike traditional methods, our approach does not rely on imposed geometric models or Fourier filtering. Instead, we track recursive coherence via golden-dimensional embedding, slippage detection, and ϕ-based persistence scoring. Recursive alignment governs whether information or energy is retained or lost across patterned systems.

1 Introduction

Patterned media are spatially structured materials with designed electromagnetic, acoustic, or quantum properties. Classical models typically treat such systems through Bloch wave expansions or boundary scattering. R-TFT offers a fundamentally different perspective: persistence is not defined by field overlap or impedance matching, but by recursive coherence alignment across space and time.

2 Recursive Embedding of Pattern Scale

Let r denote the characteristic length scale or spacing of pattern units. Define:

$$\mathrm{Dim}(r) = \frac{\ln r}{\ln \phi} \tag{1}$$

The golden-dimensional embedding reveals whether a pattern resonates within ϕ-recursive bands.

2.1 Coherence Deviation and Score

$$C_\phi(r) = |\mathrm{Dim}(r) - \mathrm{round}(\mathrm{Dim}(r))| \tag{2}$$

$$S_\phi(r) = 1 - C_\phi(r) \tag{3}$$

If $S_\phi > \phi^{-1}$, the pattern maintains recursive coherence across layers or wavefronts.

3 Real-Time Recursive Field Tracking

For an incident wave with phase tracking $\dot{\theta}(t)$ and projected direction P:

$$R(t) = \frac{\dot{\theta}(t) \cdot P}{\|P\|} \tag{4}$$

$$R_{\text{clean}}(t) = 2R_{\text{inner}}(t) - R_{\text{outer}}(t) \tag{5}$$

This filtered phase measure detects recursive drift in surface interactions, interface losses, and coherence echoes.

4 Patterned Media Persistence Criterion

A patterned field or lattice region is deemed stable if:

$$C_\phi(r) < \phi^{-1} \quad \text{and} \quad R_{\text{clean}}(t) > 0.618 \tag{6}$$

This dual filter allows real-time judgment on whether energy will localize, diffract, or re-cohere.

5 Case Studies

5.1 Fractal Photonic Lattices

When pattern unit spacing r satisfies $\text{Dim}(r) \in \mathbb{Z}$, the field retains multi-layer coherence. R-TFT predicts this without needing Bloch conditions.

5.2 Metamaterial Interfaces

Resonance bands emerge when $C_\phi(r)$ stays bounded despite geometric curvature. Recursive slippage reveals weak points not evident in impedance models.

6 Conclusion

Patterned media do not persist due to periodicity, but due to recursive alignment across ϕ-projected fields. R-TFT offers a universal method to classify coherence within any structured material or system.

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